HERIOT-WATT UNIVERSITY

F79MB Statistical Models B **Assessment Project 2 – 2026**

Your report for this project should be submitted through Turnitin by **3:30pm (local time at your campus) on Wednesday 8th April 2026**. A link to the submission page is available through 'Project 2 (60%)' in Modules or 'Assignments' on the course Canvas page. Please use the submission link provided to submit your work.

This project will account for **60%** of your grade for the course.

The total length of your report should be no more than 17 pages (11-point font, A4 size), including graphs and tables but excluding any appendices. Project title and student identification information should be in page 1, not a separate title page, and counts towards the 17-page limit. Please note that the **overall exposition/presentation is worth 10 marks** for this project.

You may consult with your lecturer or colleagues but your report must be your own work. **Plagiarism** is a serious academic offence and carries a range of penalties, some very serious – see Appendix A. You are required to submit a Standard Declaration of Authorship form. The form is available through 'Project 2 (60%)' on the course Canvas page. **You must sign this Declaration of Authorship for Project 2 before you can have access to the project 2 questions**.

- Answer all tasks in this project, making sure that you add appropriate context and that your answers have a clear and logical structure and are well presented.
- You should explain carefully your work for each task so that your work demonstrates understanding of the methodology and computations that you use.
- You should include clearly labelled and correctly referenced graphs and add appropriate comments.
- You should use `R` to perform the required analyses and produce suitable graphs. Unless otherwise stated you do not need to explain the `R` commands that you use. However you must put the `R` code in appropriate appendices at the end of your report, and should include appropriate commenting within your `R` code.
- See Appendix B for the marks allocated to overall exposition/presentation.
- **Late project submissions will be penalised according to the University Policy on Submission of Coursework. That is, work submitted after the deadline but within 5 working days will be subject to a 30% deduction from the mark awarded; work submitted more than 5 working days after the deadline will be awarded a mark of zero. No individual extensions are permitted. In the case where a student submits coursework up to five working days late and has valid mitigating circumstances, the mitigating circumstances policy will apply.**

You can expect to receive feedback on your work by 6th May 2026.

Tasks:

1. The data in the `MobileNetwork.txt` file, available in the Project 2 folder on Canvas, contains information on factors affecting mobile data speed experienced by students at Heriot-Watt University while using the campus mobile network (e.g. during lectures, online learning, social media usage, and video streaming). The dataset consists of a sample of 200 network observations collected under different network conditions across the university campus.

   You can load the data into `R` using a command like `read.table("C:/R/MobileNetwork.txt", header = TRUE)`. Note that you will need to specify your own file path.

   The recorded variables are:

   - `signal` = mobile signal strength (in dBm)
   - `bandwidth` = available network bandwidth (in MHz)
   - `legacy_users` = number of users connected using legacy mobile technologies (e.g. 2G/3G)
   - `speed` = achieved mobile data speed (in Mbps)

   Signal strength is measured in dBm, which is a logarithmic scale referenced to 1 mW. Since the received power in mobile networks is usually below 1 mW, the corresponding dBm values are negative, with values closer to zero representing stronger signals.

   It is proposed to investigate the relationship between `speed` and the explanatory variables `signal`, `bandwidth`, and `legacy_users`.

   (a) Produce an appropriate plot showing the pairwise relationships between the response variable `speed` and the explanatory variables `signal`, `bandwidth`, and `legacy_users`. Comment on your results.

   [3 marks]

   (b) Fit a multiple linear regression model with `speed` as the response variable and `signal`, `bandwidth`, and `legacy_users` as explanatory variables, excluding interaction terms. Carry out appropriate analyses to decide which terms should be retained in the model.

   Your report should include:

   - appropriate `R` output,
   - diagnostic plots,
   - clear comments on model assumptions,
   - comprehensive conclusions based on the analysis, and
   - suggestions for improving the fitted model.

   [8 marks]

   (c) In an effort to improve the analysis in part (b), fit a multiple linear regression model with `speed` as the response variable and `bandwidth`, its quadratic term, `signal`, and the interaction between the quadratic term and `signal`. The proposed model can be expressed in `R` as:

$$\texttt{speed} \sim \texttt{bandwidth} + I(\text{bandwidth}^2) * \texttt{signal}$$

Carry out appropriate analyses to determine which terms should be retained in the model. Your report should include relevant R output, plots, comments, and a thorough model diagnostic analysis.

[8 marks]

(d) Of the models fitted in parts (b) and (c), which model would you prefer for explaining mobile data speed on the Heriot-Watt University campus? Justify your answer clearly.

[3 marks]

2. The dataset in the `rent.xlsx` file, available on Canvas, contains daily bike-sharing (bike rental) data. It consists of 366 observations collected over the year 2024. The goal of this study is to construct a generalized linear model (GLM) to forecast the number of casual bike users (non-registered renters), taking into account daily and weather-related factors.

You can load the data into R using a command like `rent <- read.table("rent.csv", header = TRUE, sep = ",")`. Note that you will need to specify your own path to the file.

The variables given are:

- `casual` = number of casual bike rentals (non-registered users)
- `temp` = normalized daily temperature
- `season` = season (1 = spring, 2 = summer, 3 = fall, 4 = winter)
- `holiday` = whether the day is a public holiday (0 = no, 1 = yes)
- `workingday` = whether the day is a working day (0 = no, 1 = yes)

Note that `season`, `holiday`, and `workingday` are categorical variables.

It is proposed to model `casual` (daily number of casual bike rentals) as the response variable in a Poisson GLM, using explanatory variables such as `temp`, `season`, and others selected from the dataset.

(a) Present appropriate numerical and graphical summaries of the distributions of the variables `casual`, `temp`, and `season` separately. Comment on your results.

[3 marks]

(b) Produce appropriate plots to explore the relationship between the response variable `casual` and the explanatory variables: (i) `season`, and (ii) `temp`, distinguished by `season`. Comment on the patterns observed in these plots.

[3 marks]

(c) Carry out an appropriate test of independence to investigate whether `holiday` is dependent on: (i) `season`, and (ii) `workingday`. Your report should include a contingency table, the corresponding p-value, and a clear conclusion.

[3 marks]

(d) Fit a Poisson GLM to model `casual`, incorporating `temp`, `season`, and their interaction. Perform appropriate analyses to determine which terms should be retained in the model. Include `R` output, plots, model checking, comments, the final fitted model, and conclusion, as well as a discussion on the adequacy of the model and potential enhancements or further analyses.

[10 marks]

(e) Refit the model from part (d) using a Negative Binomial GLM (using `glm.nb()` from the `MASS` package). Provide `R` output, plots, and model diagnostics. Compare this Negative Binomial model with the standard Poisson model previously fitted, discuss the selection of the preferred model and its adequacy, and provide comprehensive conclusions.

[7 marks]

(f) Explain why it is not appropriate to include both `holiday` and `workingday` as explanatory variables in the same model, whether using a Poisson or a negative binomial model, for predicting casual bike rentals. Support your answer using the structure of the data and an appropriate statistical test.

[2 marks]

[Overall exposition/presentation: 10 marks]

[Project total: 60 marks]

[END OF PROJECT]

Appendix A: Plagiarism

- Coursework reports must be written in a student's own words and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
- Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.
- Students must never give hard or soft (electronic) copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
- Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- See `https://www.hw.ac.uk/students/studies/examinations/plagiarism.htm` for more details.

Appendix B: Rubric for marks allocated to overall exposition/presentation

The 10 marks available for the overall exposition/presentation of your report will be awarded according to the scale below.

3 marks	• Some answers lack clarity and logical structure • Some analyses may not be relevant • Statements of conclusions are present but may not be very clear • Some statistical calculations and methodology set out for the reader • Some tables and figures relevant and referenced
5 marks	• Answers sometimes clearly and logically structured • Focus on key points but may include some superfluous/irrelevant analyses • Statements of conclusions generally suitable for a non-statistician • Statistical calculations and methodology are generally presented okay for the reader • Tables and figures mostly well chosen, clear, and correctly referenced • Some sources are clearly and correctly referenced, with R code included
8 marks	• Answers are mostly clear and logically structured • Clear focus on key points, avoiding superfluous/irrelevant analyses • Statements of conclusions suitable (wherever possible) for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures well chosen, clear, correctly referenced and easy to interpret • Sources used clearly and correctly referenced • R code included with clear comments
10 marks	• As in the previous category, but the answers are very clear, strong, and logically structured. Additionally, the work demonstrates mathematical sophistication and insight, with evidence of original thought and reasoning. It also maintains a highly focused approach on key points, without superfluous or irrelevant analyses. R code is included with clear comments and without superfluous coding.